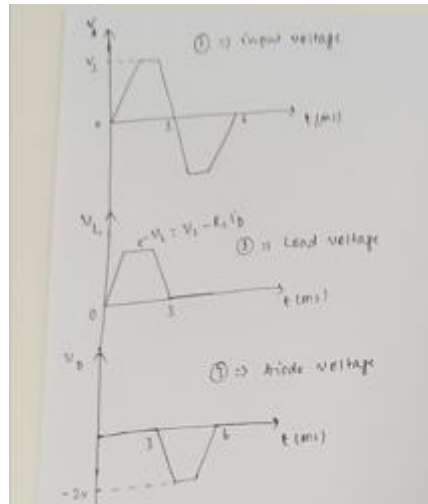


Tutorial 8 Answer Key

1. (a) 3.95 mA
(b) 0.13 mA
(c) 1 mA

2.

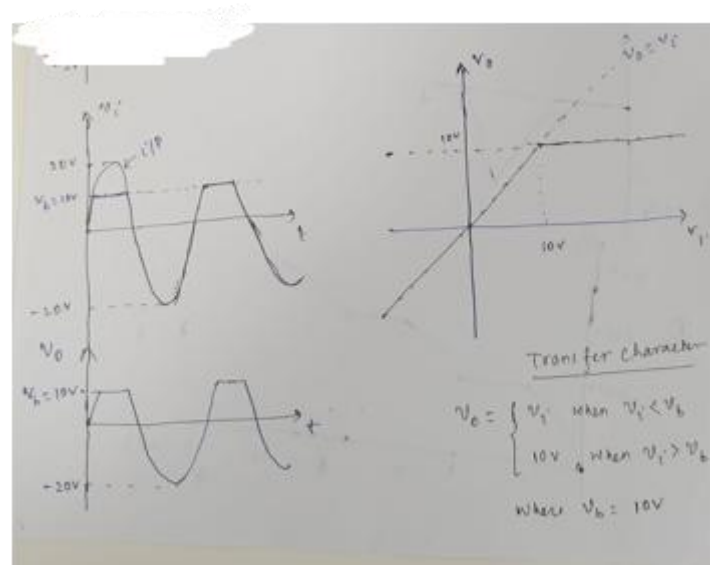


3. (a)

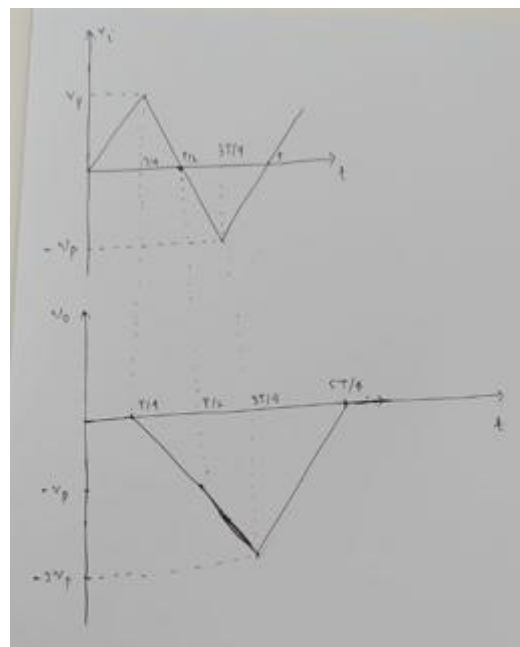
V_A (Volt)	I_A (mA)	I_B (mA)
-1	5.4	0
-0.1	4.5	0
1	0	4.4
2	0	4.4

(b) $I_A = 2.4 \text{ mA}$ $I_B = 1 \text{ mA}$

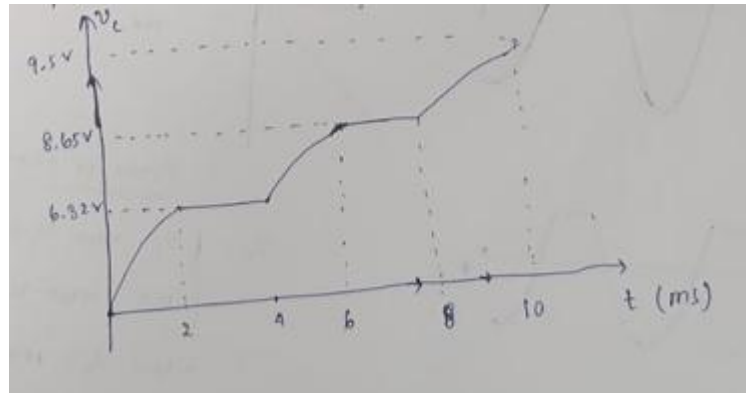
4.



5.



6.



7. $V_2 = \begin{cases} 0, & -20 \leq V_1 < 0 \\ 0.67V_1, & 0 \leq V_1 < 15 \\ 10, & 15 \leq V_1 \leq 20 \end{cases}$ Volts

8. $R_s = 2.62\Omega$